

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Fundamentals of Data Science **COURSE CODE:** DSA04

COURSE OUTCOME COVERED

CO4: Develop machine learning models for classification, regression, and clustering, including performance evaluation and methodology comparison. (BL3)

Assessment Tool Weightage: 30%

QUESTIONS: 01 Total Marks:100 Annexure A

Industry Problem Solving

Code of Conduct:

I Anto Chriswin G (192324093) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



RUBRICS FOR CALCULATION

SCENARIO BASED ASSESSMENT							
Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Level	Marks
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario		
Identification of Key Issues	20%	Accurately identifies all relevant issues	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues		

		and factors involved					
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts		
Decision Making	20%	Makes well reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided		
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand		

Question 7: Healthcare Treatment Cost Prediction Using Lasso Regression

A hospital wants to predict **treatment cost** based on patient health parameters. Since many medical features are available, the hospital wants to use Lasso Regression to identify important predictors.

Patient Age BP Sugar Level BMI Previous Visits Treatment Cost

P1 25 115 90 22 1 5000

P2 35 125 110 25 2 8000

P3 45 140 150 29 3 15000

P4 55 155 180 32 5 25000

P5 60 165 200 35 6 32000

P6 30 120 100 24 1 7000

P7 50 150 170 31 4 22000

P8 40 135 130 27 2 12000

Tasks

- a) Create the dataset using Python.
- b) Identify features and target variable.
- c) Split into training and testing sets.
- d) Implement **Lasso Regression**.
- e) Try alpha values such as **0.1, 1.0, and 10.0**.
- f) Predict treatment cost for test patients.
- g) Predict treatment cost for a new patient:

Age BP Sugar Level BMI Previous Visits

48 145 160 30 4

- h) Evaluate using MAE, MSE, RMSE, and R2 score.
- i) Print the model coefficients.
- j) Explain which medical features are important based on Lasso coefficients.

Code :

```
1 # Question 7: Healthcare Treatment Cost Prediction Using Lasso Regression
2
3 # -----
4 # 1. Import required libraries
5 # -----
6 import pandas as pd
7 import numpy as np
8
9 from sklearn.model_selection import train_test_split
10 from sklearn.preprocessing import StandardScaler
11 from sklearn.linear_model import Lasso
12 from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
13
14
15 # -----
16 # 2. Create the dataset
17 # -----
18 data = {
19     'Patient': ['P1', 'P2', 'P3', 'P4', 'P5', 'P6', 'P7', 'P8'],
20     'Age': [25, 35, 45, 55, 60, 30, 50, 40],
21     'BP': [115, 125, 140, 155, 165, 120, 150, 135],
22     'Sugar_Level': [90, 110, 150, 180, 200, 100, 170, 130],
23     'BMI': [22, 25, 29, 32, 35, 24, 31, 27],
24     'Previous_Visits': [1, 2, 3, 5, 6, 1, 4, 2],
25     'Treatment_Cost': [5000, 8000, 15000, 25000, 32000, 7000, 22000, 12000]
26 }
27
28 df = pd.DataFrame(data)
29
30 print("Original Dataset:")
31 print(df)
```

```

1 # -----
2 # 3. Identify input features and target variable
3 # -----
4
5 # Input / Independent variables
6 X = df[['Age', 'BP', 'Sugar_Level', 'BMI', 'Previous_Visits']]
7
8 # Output / Dependent variable
9 y = df['Treatment_Cost']
10
11 print("\nInput Features:")
12 print(X.columns.tolist())
13
14 print("\nTarget Variable:")
15 print(y.name)
16
17
18 # -----
19 # 4. Train-Test Split
20 # -----
21
22 # 75% training and 25% testing
23 X_train, X_test, y_train, y_test = train_test_split(
24     X, y,
25     test_size=0.25,
26     random_state=42
27 )
28
29 print("\nTraining Data:")
30 print(X_train)
31
32 print("\nTesting Data:")
33 print(X_test)
34

```

```

1 # -----
2 # 5. Feature Scaling
3 # -----
4 # Scaling is important for Lasso because L1 regularization
5 # depends on the magnitude of the coefficients.
6
7 scaler = StandardScaler()
8
9 X_train_scaled = scaler.fit_transform(X_train)
10 X_test_scaled = scaler.transform(X_test)
11
12
13 # -----
14 # 6. Test different alpha values
15 # -----
16
17 alpha_values = [0.1, 1.0, 10.0]
18
19 results = []
20
21 for alpha in alpha_values:
22
23     # Create Lasso model
24     model = Lasso(alpha=alpha, max_iter=10000)
25
26     # Train model
27     model.fit(X_train_scaled, y_train)
28
29     # Predict test data
30     y_pred = model.predict(X_test_scaled)
31
32     # Evaluation metrics
33     mae = mean_absolute_error(y_test, y_pred)
34     mse = mean_squared_error(y_test, y_pred)
35     rmse = np.sqrt(mse)
36     r2 = r2_score(y_test, y_pred)
37
38     results.append([
39         alpha,
40         mae,
41         mse,
42         rmse,
43         r2
44     ])
45

```

```

1 # -----
2 # 7. Display model comparison
3 # -----
4
5 results_df = pd.DataFrame(
6     results,
7     columns=['Alpha', 'MAE', 'MSE', 'RMSE', 'R2 Score']
8 )
9
10 print("\nLasso Model Comparison:")
11 print(results_df.to_string(index=False))
12
13
14 # -----
15 # 8. Select the best alpha
16 # -----
17 # Highest R2 is considered better.
18
19 best_alpha = results_df.loc[
20     results_df['R2 Score'].idxmax(),
21     'Alpha'
22 ]
23
24 print("\nBest Alpha:", best_alpha)

```

```

1 # -----
2 # 9. Train final Lasso model using best alpha
3 # -----
4
5 final_model = Lasso(
6     alpha=best_alpha,
7     max_iter=10000
8 )
9
10 final_model.fit(X_train_scaled, y_train)
11
12
13 # -----
14 # 10. Predict treatment cost for test patients
15 # -----
16
17 y_test_pred = final_model.predict(X_test_scaled)
18
19 prediction_df = X_test.copy()
20
21 prediction_df['Actual Cost'] = y_test.values
22 prediction_df['Predicted Cost'] = y_test_pred
23
24 print("\nTest Dataset Predictions:")
25 print(prediction_df)

```

```

1 # -----
2 # 11. Evaluate final model
3 # -----
4
5 mae = mean_absolute_error(y_test, y_test_pred)
6 mse = mean_squared_error(y_test, y_test_pred)
7 rmse = np.sqrt(mse)
8 r2 = r2_score(y_test, y_test_pred)
9
10 print("\nFinal Model Evaluation:")
11 print("MAE :", mae)
12 print("MSE :", mse)
13 print("RMSE :", rmse)
14 print("R2 Score:", r2)
15
16
17 # -----
18 # 12. Predict treatment cost for a new patient
19 # -----
20
21 new_patient = pd.DataFrame({
22     'Age': [48],
23     'BP': [145],
24     'Sugar_Level': [160],
25     'BMI': [30],
26     'Previous_Visits': [4]
27 })
28
29 # Scale new patient using the same scaler
30 new_patient_scaled = scaler.transform(new_patient)
31
32 # Prediction
33 new_prediction = final_model.predict(new_patient_scaled)
34
35 print("\nNew Patient:")
36 print(new_patient)
37
38 print("\nPredicted Treatment Cost for New Patient:")
39 print("$", round(new_prediction[0], 2))

```

```

1 # -----
2 # 13. Print Lasso coefficients
3 # -----
4
5 coefficients = pd.DataFrame({
6     'Feature': X.columns,
7     'Coefficient': final_model.coef_
8 })
9
10 print("\nLasso Model Coefficients:")
11 print(coefficients)
12
13
14 # -----
15 # 14. Identify important features
16 # -----
17
18 print("\nFeature Importance Based on Lasso Coefficients:")
19
20 for feature, coefficient in zip(X.columns, final_model.coef_):
21
22     if coefficient != 0:
23         print(
24             feature,
25             "-> Important",
26             "(Coefficient =", round(coefficient, 2), ")"
27         )
28     else:
29         print(
30             feature,
31             "-> Removed by Lasso"
32         )
33
34
35 # -----
36 # 15. Final interpretation
37 # -----
38
39 print("\nInterpretation:")
40 print("Lasso Regression uses L1 regularization to reduce")
41 print("the effect of less important features.")
42 print("Features with coefficients close to or equal to zero")
43 print("have less influence on treatment cost.")
44 print("Features with larger absolute coefficients have")
45 print("greater influence on the predicted treatment cost.")

```

Output:

Original Dataset:

	Patient	Age	BP	Sugar_Level	BMI	Previous_Visits	Treatment_Cost
0	P1	25	115	90	22	1	5000
1	P2	35	125	110	25	2	8000
2	P3	45	140	150	29	3	15000
3	P4	55	155	180	32	5	25000
4	P5	60	165	200	35	6	32000
5	P6	30	120	100	24	1	7000
6	P7	50	150	170	31	4	22000
7	P8	40	135	130	27	2	12000

Input Features:

['Age', 'BP', 'Sugar_Level', 'BMI', 'Previous_Visits']

Target Variable:

Treatment_Cost

Training Data:

	Age	BP	Sugar_Level	BMI	Previous_Visits
0	25	115	90	22	1
7	40	135	130	27	2
2	45	140	150	29	3
4	60	165	200	35	6
3	55	155	180	32	5
6	50	150	170	31	4

Testing Data:

	Age	BP	Sugar_Level	BMI	Previous_Visits
1	35	125	110	25	2
5	30	120	100	24	1

Lasso Model Comparison:

Alpha	MAE	MSE	RMSE	R2 Score
0.1	1013.608190	2.047084e+06	1430.763335	-7.188335
1.0	1013.238021	1.979174e+06	1406.831044	-6.916694
10.0	1245.618552	1.877950e+06	1370.383010	-6.511798

Best Alpha: 10.0

Test Dataset Predictions:

	Age	BP	Sugar_Level	BMI	Previous_Visits	Actual Cost	Predicted Cost
1	35	125	110	25	2	8000	8674.318268
5	30	120	100	24	1	7000	5183.081163

Final Model Evaluation:

MAE : 1245.618552478998

MSE : 1877949.5929575725

RMSE : 1370.3830095843907

R2 Score: -6.51179837183029

New Patient:

	Age	BP	Sugar_Level	BMI	Previous_Visits
0	48	145	160	30	4

Predicted Treatment Cost for New Patient:

₹ 19932.89

Lasso Model Coefficients:

	Feature	Coefficient
0	Age	-5931.670409
1	BP	9345.594688
2	Sugar_Level	-0.000000
3	BMI	0.000000
4	Previous_Visits	5439.134059

Feature Importance Based on Lasso Coefficients:

Age -> Important (Coefficient = -5931.67)

BP -> Important (Coefficient = 9345.59)

Sugar_Level -> Removed by Lasso

BMI -> Removed by Lasso

Previous_Visits -> Important (Coefficient = 5439.13)

Interpretation:

Lasso Regression uses L1 regularization to reduce the effect of less important features.

Features with coefficients close to or equal to zero have less influence on treatment cost.

Features with larger absolute coefficients have greater influence on the predicted treatment cost.